

Liquidity and rate changes: What will happen to legacy contracts?

European benchmark reforms

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Rabobank

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Huib Verbeek heads the Relative Value Trading team at Rabobank Markets.

As a Trading Head he is responsible for the strategic trading portfolio of Rabo Markets in rates products. The trading portfolio includes non client related relative value strategies in government bonds, covered bonds Supra- and Agency bonds, interest rate derivatives and listed options on futures across the curve. The scope of the traded issuance is investment grade. Before joining the Rabobank he worked at ING financial Markets heading different trading teams within the fixed income group., the last 6 years as Head of Trading Financial Markets in Amsterdam. As such he has a broad understanding of the fixed income industry and global financial markets. He holds a Master's degree in International Law at the University of Groningen.

Critical issues and questions



- Preparing for the zero liquidity scenario (slides #19,20)
- What happens to the underlying contracts once EONIA is removed (slides #12,13)?
- Is there time to build sufficient liquidity before engaging in trading activity? (slides #6,8,10,23)
- How should banks prepare for the possibility of the number of contributors to Libor dropping significantly without triggering fallback clauses (slides #15,16,22,23,24) ?

Agenda



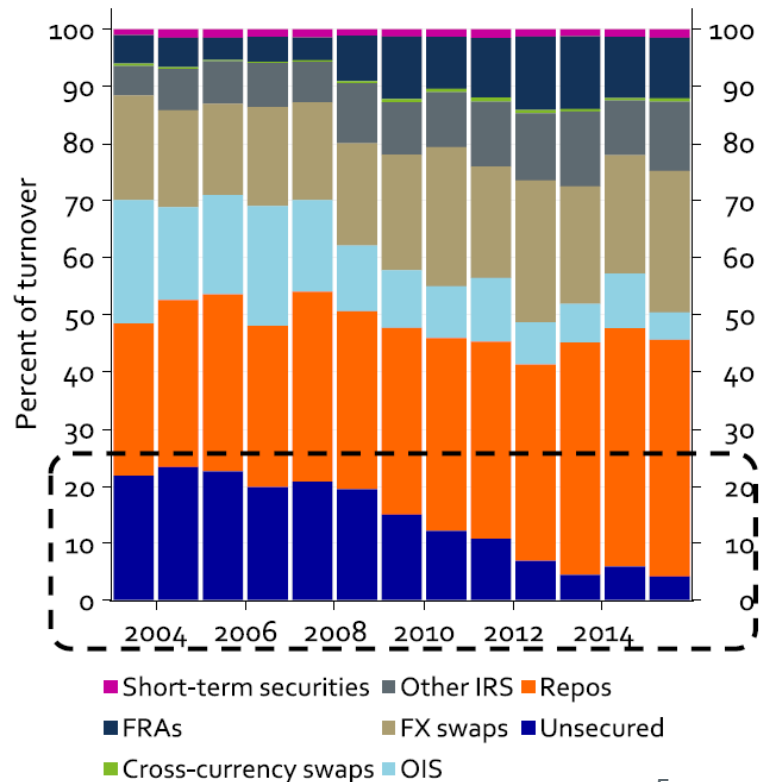
1. Background and status quo – EUR, GBP, USD
2. Scenarios for EONIA
3. Scenarios for EUIRBOR/LIBOR
4. Preparation for “zero liquidity” scenario – early transition
5. Liquidity and related issues – implications

1. Background and status quo



Benchmark regulation EUR

- European initiative to improve benchmarks
- **Objectives: accuracy, robustness, transparency**
- Benchmarks should be based on transactions
- BMR affects European supervised entities
- Interbank market dried up
- Affected benchmarks: EONIA, EURIBOR
- Related initiatives internationally: USD Libor replacement with SOFR,
- GBP Libor with reformed SONIA, CHF TOIS to SARON etc.



1. Background and status quo – US and UK

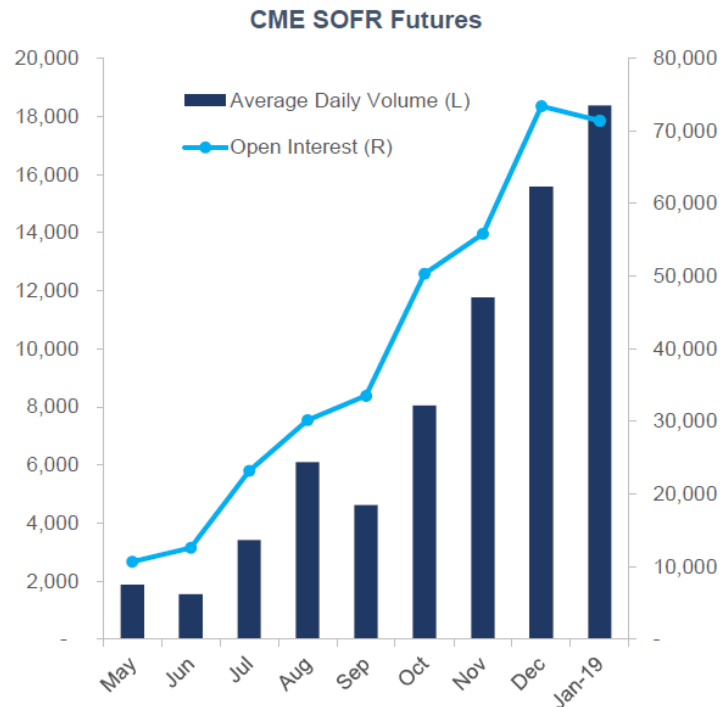


USD Libor transition

- 2021 planned termination of LIBOR (thin volume, few transactions)
- Fed funds rate inadequate as benchmark, **SOFR** chosen as benchmark replacement
- SOFR published since April 2018
- 1M and 3M SOFR futures trading on exchange (CME)
- SOFR basis vs. Eurodollar and/or Fed Funds available

GBP Libor transition

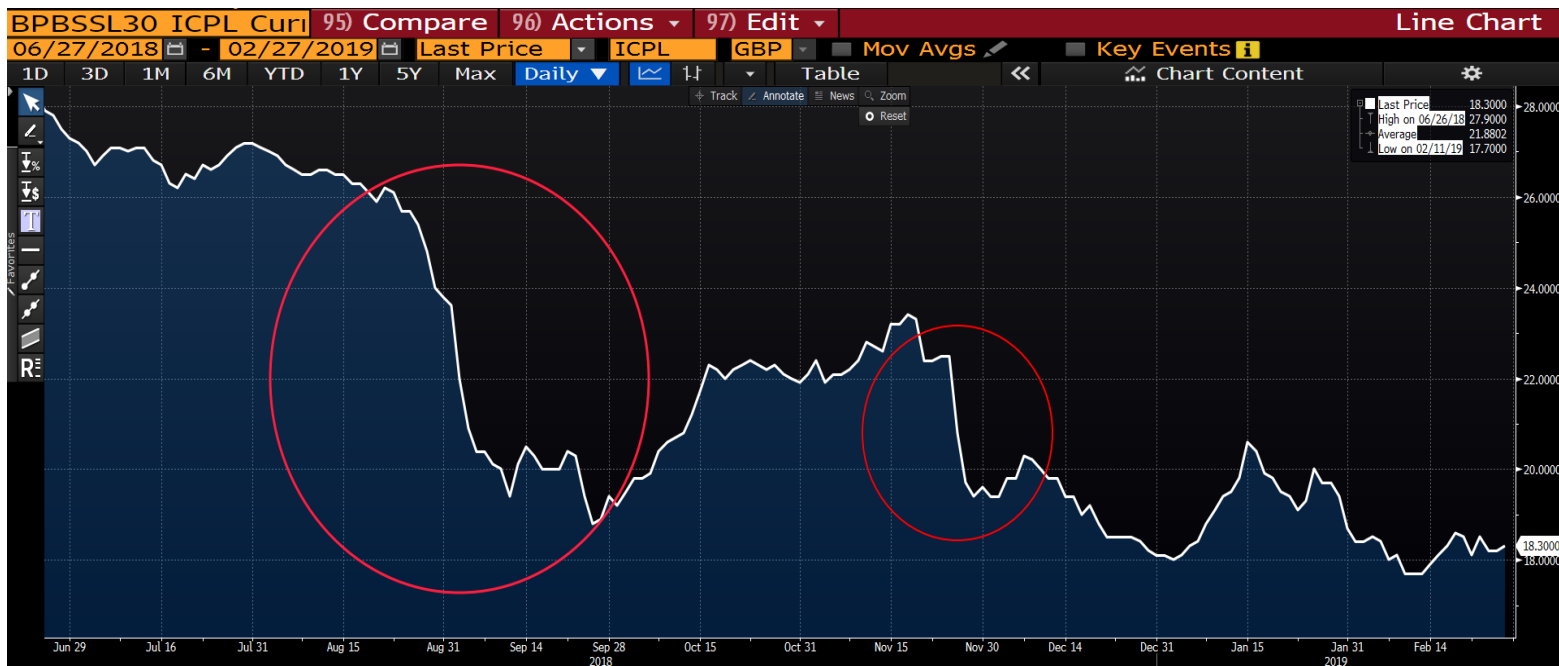
- 2021 planned termination of LIBOR (thin volume, few transactions)
- Reformed SONIA published since 2018, benchmark compliant and already in use



1. Background – market impact

GBP: “Fallback uncertainty” driving 30Y Sonia/BOR basis in second half of 2018

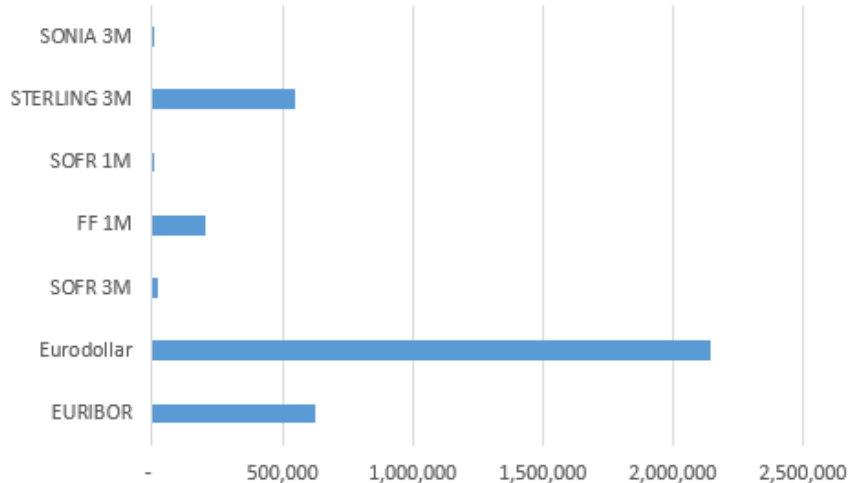
- Market consensus that “single rate fallback solution” will be result of ISDA consultation emerges.
- Minutes of UK Working Group on Sterling RFR on 31 August.
- Preliminary and final results of ISDA consultation on 27 November and 20 December 2018 respectively.
- Move in GBP (basis tightening by 7bps Aug-Sep) spilling over into 30Y EONIA/BOR, but by a far smaller extent (3bps for same period).



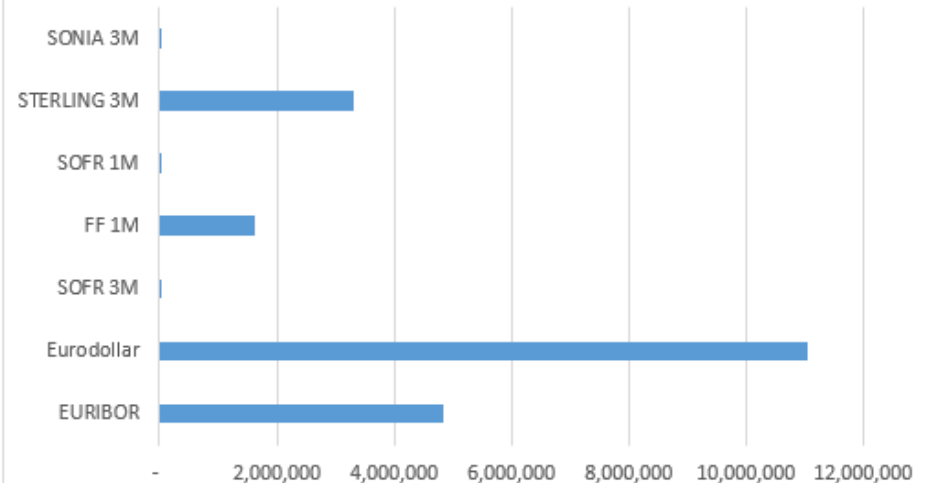
1. Background and status quo – Volumes and Open Interest compared



Cumulated Average Volume 30D (1 March 2019)



Cumulated Open Interest (1 March 2019)



Source: Bloomberg

1. Background and status quo – EUR BMR

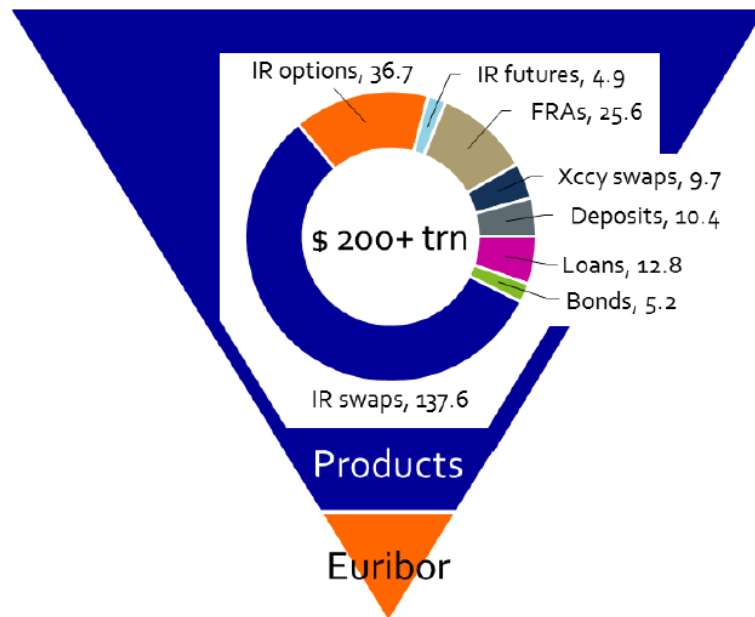


EURIBOR and EONIA transition

- *ESTER publishing to start October 2019*
- *End of transition period 31.12.2020 -> extended by 2 years on 26 February 2019*
- *Hybrid EURIBOR not yet compliant*
- *EMMI to apply for authorization only by Q2 2019*

Composition of outstanding/affected instruments

- Derivatives: swaps, futures, forwards, non-linear derivatives (caps, floors etc)
- Floating rate notes
- Bank loans based on Euribor
- Deposits, mortgages based on floating rate/Euribor



Note: Diagram not to scale, for illustrative purpose only
Source: Financial Stability Board, 2014

1. Background and status quo

Europe is late to the party

<i>Benchmark</i>	<i>Replacement</i>	<i>Secured?</i>	<i>Term struct.?*</i>	<i>Status</i>
USD Libor	SOFR	Yes (repo)	Overnight	Available
GBP Libor	SONIA+	No	Overnight	Available
CHF Libor	SARON	Yes (repo)	Overnight	Available
JPY Libor	TONAR	No	Overnight	Available
Eonia	Ester	No	Overnight	Ready, live from Oct. 2019
Euribor	Hybrid-Euribor (or Ester?)	No	Yes (hybrid-Euribor)	Testing & feedback phase 'ready in time for BMR'
EUR Libor	Ester?**	No	Overnight	Awaiting Ester (?)

* Term structure available in the fixing; this does not account for derivative solutions for a new term structure.

** [Ester] is motivated in the main by needing to have robust alternatives to Euribor and EONIA, but it is helpful in the context of the less used Euro LIBOR too. Once this euro RFR is in place, all 5 of the remaining LIBOR currencies will have a robust RFR alternative.

- Andrew Bailey, FCA

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2. Scenarios for EONIA replacement



Eonia will not evolve to compliance with BMR

- Post-1 January 2022, Eonia will only be available for use in existing contracts, subject to a decision by the FSMA if the conditions in art. 51(4) BMR are met
- Post-1 January 2022, new contracts need alternative RFR i.e. ESTER
- Publication of ESTER to start October 2019
- Transition period ends 1 January 2020 -> NOT ENOUGH TIME TO DEVELOP LIQUID TRADING -> **extended by 2 years!!!**

ESTER replaces EONIA

Scenario		New contracts	Legacy contracts
1	Banks continue to submit & Eonia is still published	Transition to ESTER; use in new transactions before 1 January 2022, (starting when possible)	No change: use of Eonia in legacy contracts Art. 51 (4) BMR
2	Bank stop submitting, EONIA no longer published		Transition legacy contracts to ESTER (starting when possible)

2. EONIA to ESTER - Transition



ECB working group transition approaches

- **Recalibration (preferred approach):** *fixed spread over ESTER based on e.g. historic data, no panel bank submissions necessary. Most suitable/probable to generate smooth transition*
- **Parallel run approach:** *EONIA and ESTER co-exist, no fixed relation. Two discounting regimes necessary. Drawback: spread possibly volatile, EONIA not BMR compliant.*
- **Pure succession approach:** *Eonia ceases and ESTER becomes successor on a predetermined date. Need to compensate for current spread (9bps) to avoid substantial value transfers.*
- **Contractual alternative:** *similar to the parallel run approach, but no simultaneous independent discounting regimes will coexist. EONIA contracts will remain available indefinitely but the EONIA discounting regime ceases after a discounting switch date.*

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3. Scenarios for EURIBOR replacement

Expectation: Euribor will evolve to compliance with BMR

- EMMI needs to apply for authorization by 1 January 2020/2022, expected application in Q2 2019
- Market participants need certainty w.r.t success/failure of reform process

Scenario	Outcome Euribor reform	Current Euribor continued?	New contracts	Legacy contracts
1	Hybrid Euribor	n/a - consistent with current Euribor	Smooth transition – no substantial impact	
2	Hybrid Euribor	Current Euribor stops	Transition to hybrid Euribor before 1 January 2022	Transition legacy contracts to hybrid Euribor (amendments, repapering etc.)
3.1	Hybrid Euribor NOT BMR COMPLIANT	Current Euribor still published (enough submitting banks!)	Transition of new contracts to alternative RFR (ESTER?)	No transition: use of current Euribor in legacy contracts Art. 51 (4)
3.2		Current Euribor ceases to exist (or panel banks stop submitting)		Transition of legacy contracts to alternative RFR (ESTER?; fallbacks, amendments etc.)

3. Scenarios for LIBOR

What happens to these LIBOR-based products after 2021 when

- 1) LIBOR is no longer available: fallbacks adjusted for case of discontinuance of LIBOR (not only temporary unavailability)
 - loan market
 - derivatives
 - capital markets transactions, CLOs, securitization
- 2) LIBOR remains available and enough submitting banks
- 3) LIBOR remains available but number of submitting banks decreases substantially

Scenario	Current Euribor continued?	New contracts	Legacy contracts
1.1	Current LIBORs still published (enough submitting banks!)	Transition of new contracts to alternative RFR (SONIA, SOFR)	No transition: use of current LIBOR in legacy contracts Art. 51 (4)
1.2	Current LIBORs ceases to exist (or panel banks stop submitting)		Transition of legacy contracts to alternative RFR (SONIA, SOFR; fallbacks, amendments etc.)

3. Scenarios for LIBOR – “ZOMBIE”

Scenario	Current Euribor continued?	New contracts	Legacy contracts
1.1	Current LIBORs still published (enough submitting banks!)	Transition of new contracts to alternative RFR (SONIA, SOFR)	No transition: use of current LIBOR in legacy contracts Art. 51 (4)
1.2	Current LIBORs ceases to exist (or panel banks stop submitting)		Transition of legacy contracts to alternative RFR (SONIA, SOFR; fallbacks, amendments etc.)

For legacy contracts referenced to UK, US Libor:

- Continuation assumes enough submitting banks.
- **Adverse scenario: Number of submitting banks drops significantly -> “Zombie” Libor** with no available hedge -> build up reserves for outlier scenarios?
- No hedge available if fallbacks do not apply as LIBOR swaps do not trade anymore
- Preemptive measures: switch to RFR where possible
- **If switch not possible (example: foreign unsophisticated counterparty does not want to switch), build reserves ????**

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4. Preparation for “zero liquidity” scenario on industry level



Risk management (ISDA, GFMA, FIA, EMTA paper 20. November 2018):

“[...] 3 approaches to managing the systemic risks associated with the cessation of an IBOR. Each of these 3 approaches are wholly appropriate in preparing for the prohibition of EONIA (and, potentially EURIBOR):

- 1) Adoption of alternative risk-free rates**
- 2) Development of robust fallbacks**
- 3) Amendment of existing contracts to reference alternative risk-free rate**

<https://www.isda.org/a/1hAEE/Briefing-on-the-need-to-extend-the-EU-Benchmarks-Regulation-Transition-P.pdf>

4. Preparation for “zero liquidity” scenario on bank level



- convert legacy contracts where possible (alternatives available?)
- add fallbacks where possible (ISDA 27 November 2018, ECB 21 January 2019)
- multilateral compression auctions for legacy swaps (example: Darrel Duffie, Sterling Risk Free Reference Rates Working Group Minutes of 12 December 2018)
- introduce contingency plans for material changes and/or cessation of current benchmark rates

Problem:

- “straight forward” for derivatives/professional market participants, BUT: loans, retail, syndicated loans etc.?

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5. Liquidity and related issues - thoughts



- timing: availability of ESTER as of October 2019, related trading (liquid?) and development of full term structure trading actively in market necessary. Extension of transition period by 2 years (before deadline was 1 January 2020) announced on 26 February ideally allows for parallel development of ESTER trading and run-off of EONIA contracts??
- operational issues remain (repapering)
- fallbacks: how to define spread in RFR term structure?
- UK, US Libor in legacy contracts. Number of submitting banks drops significantly -> "Zombie" Libor with no available hedge -> build up reserves for outlier scenarios?
- transition from forward looking term rate to forward looking term rate preferable compared to backward looking overnight rate
-

5. Liquidity and related issues - thoughts



Global cohesion: uniform fallbacks across cash products and derivatives to avoid

- market fragmentation
- introduction of basis risks
- less reliable hedges
- discrepancies in documentation across currencies and products
- issues in hedge accounting

Uncoordinated approaches to fallbacks which try to cater to specifics in local markets and products - e.g. Libor with ISDA fallback, Libor with a loan fallback, Libor with an FRN fallback – should be avoided by all means!

5. Liquidity and related issues - thoughts

